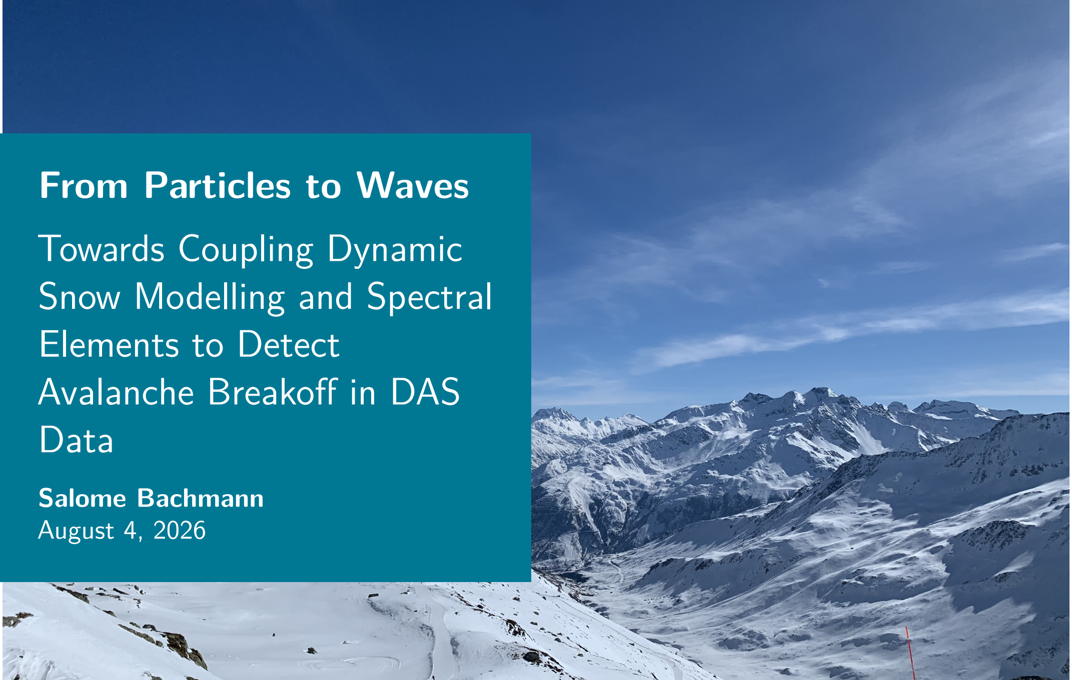

From Particles to Waves

Towards Coupling Dynamic Snow Modelling and Spectral Elements to Detect Avalanche Breakoff in DAS Data

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Overview

- Created 3 different models and 2 scenarios for crack propagation in snow using SALVUS
- Two pure spectral element (SE) models, one combined with data from material point method (MPM) simulations
- **Goal**: Couple dynamic snow modelling with SE to model crack propagation in snow

Why model crack propagation in snow? - Relevance I

- 22 fatalities by avalanche in Switzerland every year (**LongtermStatistics**)
- Avalanches are danger for human life but also infrastructure (**schweizerSnowAvalancheFormation2003**)
- Understanding processes associated with avalanches important
- Avalanche release because of loading, leading to crack propagation in a weal layer (**gaumeSnowFractureRelation2017a**)
- Crack propagation at different speeds: sub-Rayleigh (below vs of material) and Supershear (above vs) (**trottetTransitionSubRayleighAnticrack2022**), transition between two regimes possible (**bergfeldSignaturesSubRayleighSupershear2025**)
- Fast crack propagation $\rightarrow$critical crack length for release (**gaumeModelingCrackPropagation2015**) is reached faster
- Distributed Acoustic Sensing (DAS) already used to monitor avalanches in Switzerland (**paitzPhenomenologyAvalancheRecordings2023**; **edmeFiberopticDetectionSnow2023**) and Norway (**turquetAutomatedSnowAvalanche2024**)
- No knowledge if crack propagation associated with breakoff can be seen in DAS data

Bibliography I

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